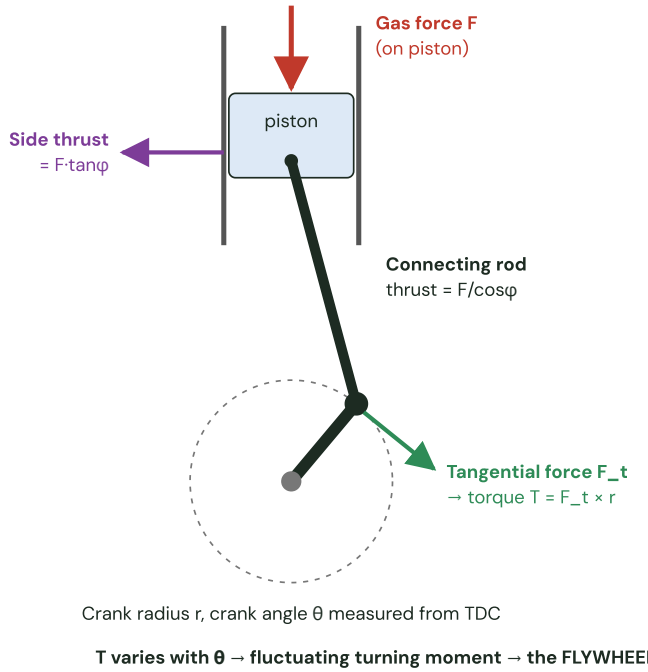
UNIT 4 · TOPIC 4(B)

Power Transmission from Piston to Flywheel

Syllabus topic: Heat load and power transmission from piston to the flywheel



Power transmission: piston → connecting rod → crankpin → flywheel (redrawn from the source figure)

Slider-crank force flow

1. Gas pressure × piston area = gas force F.
2. At the piston pin F splits into THRUST along the connecting rod = **F/cosφ** and SIDE THRUST on the cylinder wall = **F·tanφ** (the cause of piston slap and liner wear; greatest near mid-stroke).
3. At the crankpin the rod thrust resolves into a TANGENTIAL force F_t (→ torque **T = F_t × r**) and a RADIAL force (bearing load).

Crank effort: $T = F \cdot r \cdot [\sin\theta + \sin 2\theta/(2n)]$, $n = l/r$ (connecting-rod to crank ratio, typically 3.5–4.5)

Torque is ZERO at the dead centres and peaks near **25–40° ATDC** — hence the fluctuating turning-moment diagram.

Inertia forces

$$\text{Reciprocating inertia force} \approx m_{rec} \cdot r \cdot \omega^2 \cdot [\cos\theta + \cos 2\theta/n]$$

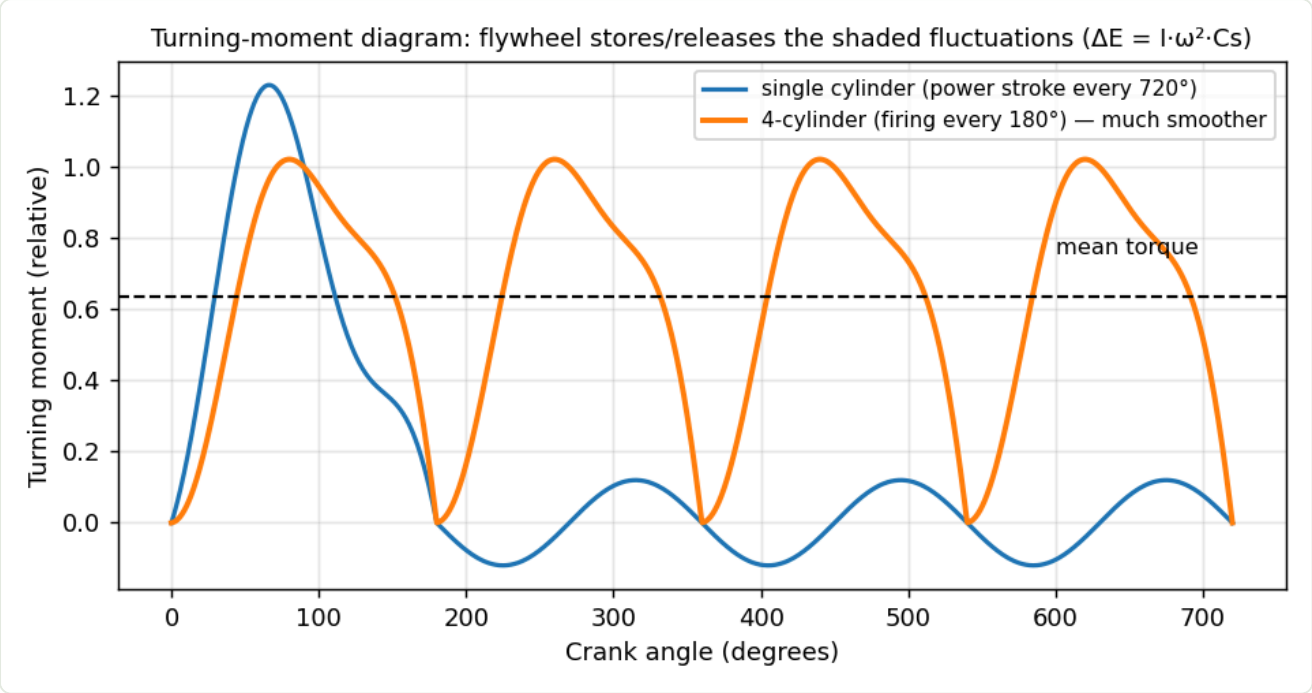
- **PRIMARY** part — frequency = crank speed. **SECONDARY** part — twice crank speed. (Balancing: Unit 2, Topic 4.)
- Rotating parts are balanced by crank webs/counterweights.

The flywheel

The flywheel stores energy during the power stroke and returns it over the other strokes, smoothing speed.

$$\text{Coefficient of fluctuation } C_s = (\omega_{max} - \omega_{min})/\omega_{mean}; \text{ required inertia } I = \Delta E/(C_s \cdot \omega^2_{mean})$$

Example. ΔE = 2000 J, C_s = 0.02, ω = 230 rad/s (2200 rpm) → I = 2000/(0.02 × 52,900) = **1.9 kg·m²**.



Turning-moment diagram — single vs 4-cylinder (indicative shapes; the mean-torque line refers to the 4-cylinder curve)

- Single-cylinder engines need the BIGGEST flywheels (720° between power strokes).
- In multi-cylinder engines the torque impulses overlap, shrinking the flywheel.

Practice MCQs

Q96. Side thrust of the piston on the cylinder wall equals:

(a) F·sinφ (b) F·tanφ
(c) F·cosφ (d) F/cosφ

Q97. Piston side thrust is the main cause of:

(a) Valve wear (b) Piston slap and liner wear
(c) Injector choking (d) Radiator scale

Q98. Crankpin torque is produced by the:

(a) Radial force (b) Tangential component of connecting-rod thrust × crank radius
(c) Gas force alone (d) Cam force

Q99. Torque from a single cylinder is ZERO at:

(a) Mid stroke (b) Dead centres
(c) 30° ATDC (d) All angles

Q100. The connecting rod-to-crank ratio n = l/r is typically:

(a) 1–2 (b) 3.5–4.5

(c) 8–10 (d) 15

Q101. Secondary inertia force has frequency:

(a) Half crank speed (b) Crank speed

(c) Twice crank speed (d) Four times

Q102. Flywheel size is LARGEST for:

(a) 6-cylinder (b) 4-cylinder

(c) Single-cylinder engines (d) All equal

Q103. Required flywheel inertia is I =

(a) $\Delta E \cdot C_s \cdot \omega^2$ (b) $\Delta E/(C_s \cdot \omega^2)$

(c) $C_s \cdot \omega^2/\Delta E$ (d) $\Delta E \cdot \omega/C_s$

Q104. ΔE = 2000 J, C_s = 0.02, ω = 230 rad/s → I ≈

(a) 0.9 (b) 1.9

(c) 3.8 (d) 9.5 kg·m²

Q105. The turning-moment diagram of a multi-cylinder engine compared with a single is:

(a) More fluctuating (b) Smoother with overlapping impulses

(c) Identical (d) Flat zero

Answers

Q96 — (b) F·tanφ

Q97 — (b) Piston slap and liner wear

Q98 — (b) Tangential component of connecting-rod thrust × crank radius

Q99 — (b) Dead centres

Q100 — (b) 3.5–4.5

Q101 — (c) Twice crank speed

Q102 — (c) Single-cylinder engines

Q103 — (b) $\Delta E/(C_s \cdot \omega^2)$

Q104 — (b) 1.9 — *2000/(0.02 × 230²) = 1.89 kg·m².*

Q105 — (b) Smoother with overlapping impulses

Sources: Liljedahl et al. — Tractors and their Power Units; standard IC-engine texts; Kepner et al.; NPTEL farm power lectures.